ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. It is crucial because it ensures data relevance, reduces computational complexity, and prevents the masking effect; where dominant patterns in large categories drown out meaningful signals in niche segments. Also, it reduces noise by excluding documents that may contain unrelated information. To add more, it helps frequent pattern mining identify words and combinations of words that are more relevant to the selected topic. Moreover, it makes the results easier to interpret because the discovered patterns can be understood within a specific subject area. Finally, isolating categories allows researchers to compare patterns across different groups and identify what makes each category distinctive. For example, in tis case, selecting the sci.med category means that the analysis focuses on medical science discussions rather than mixing them with unrelated topics such as computer graphics or religion.

AI used for this answer: Yes

If yes, how: For more explanations on FPM, to make sure that I understand them well.

External references (links): https://www.geeksforgeeks.org/dsa/frequent-pattern-mining-in-data-mining/

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

It would be a flawed data mining strategy because 40% of the documents would be lost. This would significantly reduce the amount of text available for analysis and could introduce bias if the documents with missing labels have different characteristics from the remaining documents. As a result, potentially useful information and patterns could be overlooked.

An alternative is unsupervised clustering, which groups documents according to similarities in their text without requiring predefined category labels. This allows the text from documents with missing category_name values to remain useful for analysis and helps identify patterns or topics within the dataset.

AI used for this answer: Yes

If yes, how: To learn more about the alternatives to be taken to resolve this.

External references (links): https://www.geeksforgeeks.org/data-analysis/handling-missing-values-machine-learning/

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, not perfectly, but they remain relatively the same.
2. The risk is that minority classes can become too small or disappear in the sample, causing bias and poor analysis results.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. It increases.
2. Adding stopwords decreases the overall sparsity of the matrix because common words appear in many documents. Including these words adds more non-zero entries to the document-term matrix, reducing the percentage of zeros. However, the matrix will still be sparse because most documents contain only a small fraction of all the words in the vocabulary.

AI used for this answer: Yes

If yes, how: In explaining how including stopwords affects the number of features and the sparsity of the matrix.

External references (links): https://search.brave.com/search?q=2.%09Since+stopwords+are+highly+common+words+that+appear+in+almost+all+documents%2C+what+effect+does+adding+them+back+have+on+the+overall+sparsity+%28the+percentage+of+zeros%29+of+the+matrix%3F&source=web&conversation=09ae5fb3a7074f4e348db4707a9c8b43bb79&shared=1

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. The total vocabulary size decreases as the variance threshold increases. At a threshold of 0.0, 35,482 features are kept. At 0.01, only 5,089 features remain, and at 0.05, only 1,491 features remain. This shows that increasing the threshold removes more low-variance features and reduces the dimensionality of the dataset.

2. Words that appear in almost every document or in very few documents have low variance because their occurrence does not vary much across the dataset. These words provide limited information for distinguishing documents and are among the first to be eliminated as the variance threshold increases.

3. Near-zero variance features provide little variation between documents, so they often contribute limited information for distinguishing categories. Removing them reduces the number of features, memory usage, and computational cost while potentially improving model efficiency. However, it is not guaranteed to preserve accuracy in every case, so model performance should be evaluated after filtering.

Threshold Number of terms kept (keep == True)

0.0 - 35482

0.01 - 5089

0.05 - 1491

AI used for this answer: Yes

If yes, how: In explain how increasing the threshold affects vocabulary size, document frequency, and dimensionality reduction.

External references (links): https://search.brave.com/search?q=In+terms+of+document+frequency%2C+what+kinds+of+words+have+variance+close+to+zero+and+are+eliminated+first%3F+In+terms+of+document+frequency%2C+what+kinds+of+words+have+variance+close+to+zero+and+are+eliminated+first%3F&source=desktop&conversation=09aea6fb94febc21c330eae0a8d7a2e395b0&shared=1

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?

2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?

3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. The top-correlated words include both generic words and domain-specific terminology. Words such as thanks, writes, looking, and hi are common expressions that can appear in many categories. In contrast, VGA, video, and card are more closely associated with computer graphics and hardware.
   This suggests that the correlation results capture both topic-related vocabulary and general patterns in how messages are written. Although domain-specific words can help identify the comp.graphics category, generic words may also show correlations because of differences in writing styles or document formats.
2. Pearson correlation measures the strength and direction of a linear relationship between two variables using their actual numerical values. Its coefficient ranges from −1 to +1. Spearman rank correlation measures the strength and direction of a monotonic relationship by converting the values into ranks before calculating the correlation. It can identify relationships that are consistently increasing or decreasing even when they are not linear.
   A word could have a high Spearman correlation but a significantly lower Pearson correlation when its relationship with a topic is monotonic but nonlinear. For example, a word's frequency might generally increase with the likelihood that a document belongs to a topic, but the increase may not be proportional. Spearman can capture the consistent ranking pattern, whereas Pearson may show a weaker linear relationship.
3. This would happen when the relationship is monotonic but not linear. For example, suppose a word appears: rarely in documents outside the topic, more often in topic-related documents, but with highly uneven frequencies or extreme outliers. A word might consistently rank higher in comp.graphics documents than in other documents, giving it a high Spearman correlation, because the rank ordering is strong. However, if the actual word counts vary wildly across documents, the linear relationship becomes weaker, so the Pearson correlation would be lower.

AI used for this answer: Yes

If yes, how: To interpret the correlation results and explain the mathematical differences between Pearson and Spearman correlation in text mining.

External references (links): https://www.metwarebio.com/pearson-vs-spearman-correlation-multi-omics-data-analysis/

https://chatgpt.com/s/t_6ad224856e388191922d5c6f033156f3

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

When minSup decreases from 9 to 3, the algorithm finds far more patterns (from 32 to 923) and takes more time to run (from about 0.012 s to 0.181 s). Lower support thresholds allow rarer patterns to be considered frequent, which increases the search space and computation cost.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The difference is in how patterns are selected, Top-K FAE (k) keeps the top k most frequent patterns. You specify how many patterns you want and the algorithm returns exactly that number of highest-ranking patterns. MaxFPGrowth (minSup) keeps all patterns whose support is at least the minimum support threshold. You specify the minimum frequency requirement. The number of returned patterns is not fixed; it depends on the data. k in Top-K FAE chooses a fixed number of top patterns, while minSup in MaxFPGrowth chooses all patterns above a support threshold.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are more concise because they keep only the largest frequent itemsets and remove redundant frequent subsets.

AI used for this answer: Yes

If yes, how: I used AI to help interpret the pattern mining results, compare how changing minSup and k affects runtime and pattern counts, and explain the conceptual difference between Top-K frequent patterns and maximal frequent patterns in clear academic language.

External references (links): https://ai.ncnu.edu.tw/c/bee57889-a1ea-4423-bfbe-97a2857d70e3

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?

2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. The raw X_counts matrix has tens of thousands of features, meaning each document exists in a very high-dimensional space. A normal scatter plot can only display, 2 dimensions on a 2D plot, or 3 dimensions on a 3D plot. Because of this, it is mathematically impossible to directly visualize all dimensions of the raw matrix at once. Visually, plotting thousands of dimensions on a standard scatter plot is not meaningful, since humans cannot interpret that many axes simultaneously.
2. We must first apply dimensionality reduction algorithms. These algorithms reduce the large number of features into a smaller number of dimensions (usually 2 or 3) while trying to preserve the important structure of the data for visualization and analysis.

AI used for this answer: Yes

If yes, how: To help explain why high-dimensional text vectors cannot be directly visualized in a scatter plot and to identify dimensionality reduction as the required category of algorithms before plotting.

External references (links): https://share.google/aimode/p1FXAdgHOU8CzloOz

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. t-SNE/UMAP show clearer cluster islands; PCA shows more overlap near the center.
2. PCA struggles because it is linear and cannot capture the non-linear structure of high-dimensional text data as well as t-SNE/UMAP.
3. In t-SNE, distances between clusters are not always meaningful globally, so large gaps should not be over-interpreted

AI used for this answer: Yes

If yes, how: To help interpret the PCA, t-SNE, and UMAP scatter plots and to explain the mathematical differences between linear and non-linear dimensionality reduction methods in clear academic language.

External references (links): https://share.google/aimode/RjTyH75hO8jJAcTLI

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. There are 4 binary columns created, one for each category.
2. It falsely assumes that the categories have a numerical order and meaningful distance between them. For example, the algorithm may incorrectly interpret, category 3 as being "greater than" category 1 and category 2 as being closer to 1 than to 3. But these category numbers are only labels, not real numeric values. There is no true mathematical meaning in the difference between 0, 1, 2, and 3.
3. One-hot binarization represents each category as a separate binary indicator column, so no category is treated as larger, smaller, or closer than another.

AI used for this answer: Yes

If yes, how: To help explain the difference between integer category encoding and one-hot binarization, and why one-hot encoding prevents algorithms from making false ordinal assumptions about class labels.

External references (links):

https://share.google/aimode/aH4DxWYHxyVrPtoL9

https://letsdatascience.com/blog/categorical-encoding-a-practical-guide-to-one-hot-label-and-target-methods

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